
LAXREE

HR & Salary Management Dashboard

Google Sheets Attendance Automation

Step-by-Step Setup Guide

What This Guide Covers

Create	Google Sheet with Master Employees + Daily Input sheets
Install	Google Apps Script for auto-calculations (OT, Status, Hours)
Connect	Link the sheet to your Laxree Dashboard for automatic sync
Automate	Daily attendance data flows from Sheet to Dashboard seamlessly

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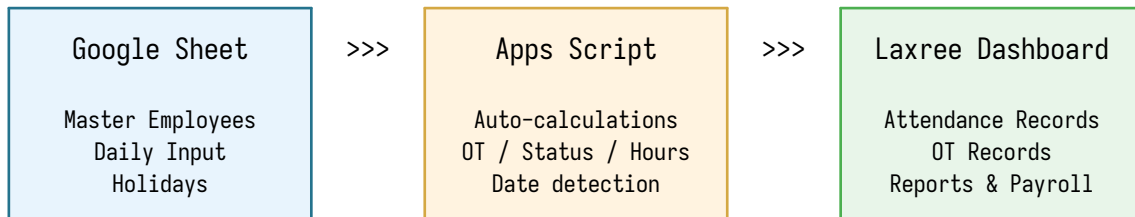
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1. Overview: How It Works

The Laxree Google Sheets Attendance Automation system creates a seamless bridge between your Google Sheet and the Laxree HR Dashboard. Instead of manually entering attendance one-by-one in the dashboard, HR staff can simply fill in In Time and Out Time in a familiar spreadsheet interface, and the data automatically syncs to the dashboard with all calculations (OT, work duration, status) handled automatically.

The system has three main components:



Master Employees Sheet: This is your single source of truth for employee data. When you add a new employee here, they automatically appear in the Daily Input sheet the next time you generate attendance rows. Fields include Employee Code, Full Name, Firm, Location, Salary Type, Shift Start/End, and more.

Daily Input Sheet: For each day, HR generates rows for all active employees. The employee names and shift details are pre-filled. HR only needs to enter In Time and Out Time. The Apps Script auto-calculates work duration, OT hours, total hours, late entry detection, half-day status, Sunday hours, and public holiday hours. The Status column is also auto-calculated using color-coded conditional formatting.

Holidays Sheet: A simple list of public holidays. When a date matches a holiday, the Daily Input sheet automatically marks employees as "holiday" and calculates PH hours.

2. Step 1: Create a Google Sheet

Start by creating a brand new Google Sheet that will serve as your attendance workbook. This sheet will contain three tabs (Master Employees, Daily Input, and Holidays) that are automatically created by the Apps Script in the next step.

Step 1: Open your web browser and go to sheets.google.com

Step 2: Click the blank (+) button to create a new spreadsheet

Step 3: Name the spreadsheet something recognizable like "Laxree Attendance 2026"

Step 4: Leave the default "Sheet1" tab as-is (it will be deleted by the script)

Step 5: Note the Sheet ID from the URL:

https://docs.google.com/spreadsheets/d/SHEET_ID_HERE/edit

Step 6: Copy and save this Sheet ID somewhere safe; you will need it later to connect to the dashboard

Important: The Sheet ID is the long string of characters in the URL between /d/ and /edit. For example, if the URL is <https://docs.google.com/spreadsheets/d/1aBcDeFgHiJkLmNoPqRsTuVwXyZ/edit>, then the Sheet ID is: 1aBcDeFgHiJkLmNoPqRsTuVwXyZ

3. Step 2: Add the Google Apps Script

The Google Apps Script is the brain of the attendance automation. It creates the sheet structure, auto-populates employee data, calculates working hours, OT, status, and provides the sync functionality with your Laxree Dashboard. Follow these steps carefully to install it:

Step 1: In your Google Sheet, click Extensions from the top menu bar

Step 2: Select Apps Script from the dropdown menu. A new tab will open with the Apps Script editor

Step 3: You will see a default file called "Code.gs" with an empty function. Delete all the existing code in the editor

Step 4: Open the file "Laxree_Attendance_Google_Apps_Script.gs" that was provided with this guide

Step 5: Copy the entire contents of that file and paste it into the Apps Script editor, replacing everything

Step 6: Click the Save icon (floppy disk) or press Ctrl+S to save the script

Step 7: You may be prompted to authorize the script. Click "Review permissions", select your Google account, click "Advanced", then click "Go to Laxree HR (unsafe)", and finally click "Allow"

Authorization Note: The script needs permission to edit your spreadsheet, access the calendar/timezone, and make HTTP requests (for syncing with the dashboard). These are standard permissions for Apps Script. The "unsafe" warning appears because the script is not published on the Google Workspace Marketplace; this is completely normal for custom scripts.

4. Step 3: Initialize the Sheet

After installing the Apps Script, you need to run the initialization function to create the sheet structure. This will set up three sheets (Master Employees, Daily Input, Holidays) with proper headers, formatting, data validation dropdowns, and

default values. This step only needs to be done once.

Step 1: In the Apps Script editor, look at the toolbar at the top. You will see a function dropdown selector

Step 2: Click the dropdown and select "initializeSheet" from the list of functions

Step 3: Click the Run button (play icon) next to the dropdown. The script will execute

Step 4: Wait for the execution to complete. You will see a progress indicator in the editor

Step 5: Switch back to your Google Sheet tab. You should now see three new sheets:

Sheet Name	Purpose	Key Columns
Master Employees	Employee database - add/edit employees	Employee Code, Full Name, Firm, Location, Salary Type, Employment Type, Active
Daily Input	Daily attendance - HR fills In/Out Times	Employee Code, Name, Date, In Time, Out Time, Work Duration
Holidays	Public holidays for PH detection	Date, Holiday Name, Type

Also note: The default "Sheet1" tab will be automatically deleted. The "Master Employees" sheet will be set as the active sheet. The header rows are styled with the Laxree brand colors (dark background, gold text) and are protected from accidental editing.

5. Step 4: Add Employees to Master Sheet

Now that the sheet is initialized, you need to add your employee data to the Master Employees sheet. This data will be used to auto-populate the Daily Input sheet. When you add new employees in the future, they will automatically be included the next time you generate daily attendance rows.

You have two options for adding employees:

Option A: Manual Entry - Simply type the employee data directly into the Master Employees sheet. Most columns have dropdown validation (Firm, Location, Salary Type, Employment Type, Active) to prevent typos. The Hourly Rate and OT Rate columns have auto-calculating formulas based on Monthly Salary and Shift Hours.

Option B: Use Sample Data - The Apps Script includes a "Add Sample Employees" function that adds 10 sample employees with realistic data. This is useful for testing before entering your real employee data.

Step 1: To add sample employees: In the sheet, click Laxree HR menu (top menu bar) then Add Sample Employees

Step 2: To enter manually: Go to the Master Employees sheet and fill in the columns starting from Row 2

Step 3: Required fields: Employee Code (Col A), Full Name (Col B), Firm (Col C), Location (Col D)

Step 4: Optional fields: Salary Type (defaults to "hourly"), Shift Start (defaults to "10:00"), Shift End (defaults to "19:00")

Step 5: Make sure the Active column (Col K) is set to "Yes" for employees who should appear in daily attendance

Step 6: The Hourly Rate (Col H) and OT Rate (Col I) are auto-calculated from Monthly Salary and Shift Hours

6. Step 5: Generate Daily Attendance

Each day, HR needs to generate attendance rows for all active employees. The Apps Script pulls the employee list from the Master Employees sheet and creates a row for each employee with their name, firm, location, shift details, and the selected date. HR then simply fills in the In Time and Out Time columns.

Step 1: In the Google Sheet, click the Laxree HR menu from the top menu bar

Step 2: Select "Generate Today's Attendance" to create rows for today, or "Generate Daily Attendance" for a custom date

Step 3: The script will read all active employees from Master Employees and create one row per employee in the Daily Input sheet

Step 4: Each row is pre-filled with: Employee Code, Name, Firm, Location, Date, Shift Start, Shift End, Shift Hours

Step 5: If today is a Sunday, all rows will be pre-marked as "weekly-off". If today is a holiday, they will be marked as "holiday"

Step 6: If attendance for the selected date already exists, you will be asked if you want to regenerate (this will clear existing In/Out times)

7. Step 6: Enter In Time / Out Time

This is the main daily task for HR. After generating the attendance rows, simply fill in the In Time and Out Time columns for each employee. All calculations happen automatically through spreadsheet formulas. The auto-calculated columns are read-only and update instantly when you enter the time values.

Column	You Fill?	Description
Employee Code	Auto	Pulled from Master Employees
Employee Name	Auto	Pulled from Master Employees
Firm / Location	Auto	Pulled from Master Employees
Date	Auto	Set when generating attendance
In Time (Col F)	YES	Enter check-in time as HH:MM (e.g., 10:00, 09:15)
Out Time (Col G)	YES	Enter check-out time as HH:MM (e.g., 19:00, 20:30)
Shift Start/End	Auto	From Master Employees per employee
Work Duration (Col K)	Auto	Calculated: (Out Time - In Time) in hours
OT Hours (Col L)	Auto	Calculated: $\max(0, \text{Work Duration} - \text{Shift Hours})$
Total Hours (Col M)	Auto	Same as Work Duration (already includes OT)
Status (Col N)	Auto	Present / Late / Half-Day / Absent / Weekly-Off / Holiday
Late Entry (Col O)	Auto	TRUE if In Time > Shift Start + 15 min grace
Half Day (Col P)	Auto	TRUE if Work Duration < Shift Hours / 2
Sunday Hours (Col Q)	Auto	Work Duration if Sunday, else 0
PH Hours (Col R)	Auto	Work Duration if Public Holiday, else 0
Remarks (Col S)	Optional	Any notes about the attendance record

Time Format: Always enter times in 24-hour format as HH:MM. For example: 09:00 for 9 AM, 13:30 for 1:30 PM, 19:00 for 7 PM, 22:00 for 10 PM. Do not use AM/PM notation.

8. Step 7: Create Google Cloud Service Account

To connect the Google Sheet with the Laxree Dashboard, you need a Google Cloud Service Account. This is a special type of Google account that allows the dashboard to securely read and write data to your Google Sheet without needing your personal Google credentials. Think of it as a robot account that only has access to the specific spreadsheet you share with it.

Step 1: Go to console.cloud.google.com and sign in with your Google account

Step 2: Create a new project (or select an existing one). Name it "Laxree HR Integration"

Step 3: In the left sidebar, go to IAM & Admin then Service Accounts

Step 4: Click "Create Service Account". Name it "laxree-sheet-access" and click Create

Step 5: Skip the optional permissions step (click Continue), then click Done

Step 6: Click the email address of the service account you just created

Step 7: Go to the Keys tab, click Add Key then Create new key

Step 8: Select JSON format and click Create. A JSON file will download to your computer

Step 9: Open the downloaded JSON file. You need two values from it:

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client_email: Looks like "laxree-sheet-access@your-project.iam.gserviceaccount.com"  
private_key: A long string starting with "-----BEGIN PRIVATE KEY-----"
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Keep this JSON file secure. It grants access to any Google Sheet shared with the service account email.

Step 10: Now enable the Google Sheets API: Go to APIs & Services then Library

Step 11: Search for "Google Sheets API" and click Enable

Step 12: The service account is now ready to use

9. Step 8: Share Sheet with Service Account

The service account needs permission to access your Google Sheet. You grant this by sharing the sheet with the service account email address, just like you would share with a colleague.

Step 1: Open your Laxree Attendance Google Sheet

Step 2: Click the Share button in the top-right corner

Step 3: Paste the service account email from the JSON file (e.g.,
laxree-sheet-access@your-project.iam.gserviceaccount.com)

Step 4: Set the permission to Editor (the dashboard needs to read and write data)

Step 5: Uncheck "Notify people" (service accounts do not have email inboxes)

Step 6: Click Send or Share

Security Note: The service account only has access to sheets explicitly shared with it. It cannot access any other Google Drive files. If you ever need to revoke access, simply remove the service account from the Share settings of the sheet.

10. Step 9: Connect Sheet to Laxree Dashboard

Now connect the Google Sheet to your Laxree HR Dashboard. This is done through the Settings page in the dashboard where you enter the service account credentials and the Sheet ID.

- Step 1: Open the Laxree HR Dashboard in your browser and log in with admin credentials
- Step 2: Navigate to Settings (gear icon in the sidebar)
- Step 3: Scroll down to the Google Sheets Integration section
- Step 4: Enter the Service Account Email (from the JSON file: client_email field)
- Step 5: Enter the Private Key (from the JSON file: private_key field). Copy the entire key including the BEGIN/END lines
- Step 6: Enter the Google Sheet ID (from the sheet URL, as noted in Step 1)
- Step 7: Click Save & Connect. The dashboard will test the connection
- Step 8: If successful, you will see a green "Connected" message with the list of sheets found

11. Step 10: Sync Data (Pull from Sheet)

Once the Google Sheet is connected, you can sync attendance data between the sheet and the dashboard. The primary workflow is: fill data in the Google Sheet, then pull it into the dashboard.

Sync Options in the Attendance Tracker page:

Button	Action	When to Use
Pull Daily Input	Reads the "Daily Input" sheet and imports it into the dashboard database	Use this after filling in the In/Out times in the Google Sheet
Push to Sheet	Exports dashboard attendance data to the "Daily Input" spreadsheet	Use this to sync data back to the Google Sheet to start using a new sheet
Create Template	Creates empty attendance rows in "Daily Input" for each active employee	Use this to create a new template for the dashboard side instead of manually adding rows
Pull (Legacy)	Reads from the old "Attendance" sheet	Only use for backward compatibility with older sheets

Daily Workflow Summary: Each day, (1) Go to Google Sheet, (2) Click Laxree HR > Generate Today's Attendance, (3) Fill in In Time and Out Time for each employee, (4) Go to Laxree Dashboard > Attendance page, (5) Click "Pull Daily Input" button. All attendance data including OT, status, and hours will be imported automatically.

12. Auto-Sync Setup (Optional)

For advanced users, you can set up automatic synchronization so the Apps Script periodically pushes data from the Google Sheet to the Laxree Dashboard without manual intervention. This uses Google Apps Script time-based triggers.

Step 1: In the Apps Script editor, update the DASHBOARD_API_URL constant at the top of the script with your dashboard URL (e.g., "https://your-app.vercel.app/api/gsheet")

Step 2: Save the script (Ctrl+S)

Step 3: In the Apps Script editor, select the function "setupAutoSync" from the dropdown

Step 4: Click Run. When prompted, enter the sync interval in hours (e.g., 1 for every hour, 2 for every 2 hours)

Step 5: The script will create a time-based trigger that automatically calls syncToDashboard() at the specified interval

Step 6: To disable auto-sync later, select "disableAutoSync" from the function dropdown and click Run

Note: Auto-sync requires the DASHBOARD_API_URL to be set correctly and the dashboard server to be accessible from Google servers. If your dashboard is hosted on a local network or behind a firewall, auto-sync may not work. In that case, use the manual "Pull Daily Input" button from the dashboard instead.

13. How Calculations Work

The attendance calculations follow the Laxree HR policy. All calculations happen automatically in the Google Sheet via formulas and are verified again when the data is pulled into the dashboard. Here is how each field is calculated:

Field	Formula / Logic	Example
Work Duration	(Out Time - In Time) converted to hours.	From 09:00 to 18:00 is 9 hours
OT Hours	max(0, Work Duration - Shift Hours). Only counts extra hours beyond shift.	1 hour extra beyond 9 hours shift.
Total Hours	Same as Work Duration (already includes OT hours calculation).	10 hours total
Late Entry	TRUE if In Time > Shift Start + 15 minutes	In at 9:20, Shift 9:00 = Late
Half Day	TRUE if Work Duration < Shift Hours / 2.	4.0 hours < 9/2 = 4.5 = Half Day
Status	Priority: Sunday > Holiday > Half-Day > Late	Present
Sunday Hours	If day is Sunday (day 0), equals Work Duration	0 hours on Sunday
PH Hours	If date is in Holidays sheet, equals Work Duration	0 hours on Public Day

Status Priority Logic:

Priority	Condition	Status	Color Code
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1 (Highest)	Day is Sunday (dayOfWeek = 0)	weekly-off	Blue
2	Date matches a holiday in Holidays sheet	holiday	Purple
3	Work Duration < Shift Hours / 2	half-day	Orange
4	In Time > Shift Start + 15 min grace	late	Yellow
5	In Time and Out Time are filled	present	Green
6 (Lowest)	No In/Out time and not Sunday/Holiday	absent	Red

14. Troubleshooting

Here are solutions to the most common issues you may encounter while setting up or using the Google Sheets attendance automation system. Most problems are related to permissions, time formatting, or configuration mismatches.

Problem	Cause	Solution
"Laxree HR" menu not appearing	Script not saved or not authorized	Extensions > Apps Script, save the script, refresh the sheet
Formulas showing #ERROR!	Time format is wrong	Enter times as HH:MM (e.g., 10:00, not 10 AM or 10:00:00)
Status column is blank	In/Out times not filled yet	Status auto-calculates only when In Time and Out Time are both entered
OT showing 0 when it should be more	Shift Hours is wrong in Master Employees	Check the Shift Hours column in Master Employees; OT = Duration - Shift Hours
Dashboard says "Not configured"	Credentials not saved	Go to Settings > Google Sheets Integration and enter all three fields
Connection test fails	Service account not set up correctly	Verify the OAuth file is correct and the sheet is shared with service account
"Pull Daily Input" shows 0	No data in Daily Input sheet	Take wrong format "Generate Today's Attendance" first and filled with data
Employee not found error during sync	Employee Code in sheet does not match	Ensure that Employee Code in the Google Sheet matches the employee's code
Late entry not detected	Shift Start not set correctly	Check the Shift Start column in Master Employees for that employee
Holiday not detected	Date format wrong in Holidays sheet	Enter dates as YYYY-MM-DD (e.g., 2026-01-26) in the Holidays sheet

For additional support, contact the Laxree HR system administrator or refer to the dashboard documentation.